Marking Rubric – Year 3 Lab Reports – 2025 v.0.5

Aims

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
1. Structure	It is not articulated what the work was aiming to achieve.	Vague aims are provided such that success in achieving those aims is open to interpretation.		Aims are provided that specify a threshold for success.		Aims are broken down into **specific, answerable** research question(s).		All the research questions are articulated in a way which <u>clearly</u> allows them to be answered **quantitatively**.	

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
2. Independent Thought (Aims)	There was no aim or research question provided.	The research question was outside the scope of physics accessible with the equipment available.		The research question was based entirely on a task given in the lab script.		The research question was an incremental variation from the lab script (e.g., probing an assumption in the lab script).		The research question was about (<u>sensible</u>) physics, demonstrating extension beyond the methods described in the lab script.		The research question was about (<u>relevant and sensible</u>) physics that is not in the suggested directions included in the lab script.	

Overall Communication

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
3. Concise transmission of information (includes evaluation of the abstract)	The content of the report is largely irrelevant and does not answer the research question.	The structure of the account means it is difficult to understand what was done. E.g., sentences contain relevant content but more than one idea; sections/paragraphs present disconnected information.		The account provides minimal information to understand what was done such that in places there remains some ambiguity. Statements are made that are missing citations.		Sufficient <u>information, figures, and citations</u> are provided to clearly understand what was done, such that the reader could **reproduce the work**.		**Only** information and figures directly relevant to answering the research question(s) using the chosen method are provided. Other information is given by relevant citations to **academic** sources. ----- It is not possible to meet this category if the page/time limit is exceeded.	

Methods

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
4. Independent Thought (Methods)	The methods were not described or were incomprehensible .	The methods and protocols used do not go beyond details provided in the lab script.		The methods follow closely to approaches described in the lab script and provide some additional detail on intermediate steps.		Required for Upper 2nd and higher: The method itself is consistent with the aim of the report and the description is sufficient that the reader can assess that it would produce data that could answer the RQs.					
						The methods follow closely to approaches described in the lab script and give additional detail that is appropriate, relevant, and necessary for the reader to interpret the results.		The methods of data collection or analysis include some methods that are extensions to those provided in the lab script, while still being a physically sensible approach to extracting quantities of interest.		Novel methods for measuring quantities not described in the lab script are given and are fully documented such the reader could easily reproduce them.	
5. **Justification** of methods used (the choice of apparatus & associated parameters & data analysis approaches)	The method used to collect data was not described.	No justification is provided for the methods used in the data collection, or the method was used because that is what the lab script said to do.		Some effort was given in justifying the methods used to collect and analyse data including **identification** of assumptions and limitations.		The choice and development of methods to collect and analyse data is justified by considering (correctly) the **effect** assumptions and limitations would have **on the measured data and results**.		The choice of method used to collect **and** analyse data is described including how it was calibrated/optimised and any assumptions that can be justified correctly based on theoretical considerations and/or apparatus limitations. At least one assumption/choice in the method is **tested** (experimentally or with a simulation) to		+ All pertinent assumptions/choices in data collection **and** analysis have been tested quantitatively and a logically consistent argument is provided to justify how the effect of the assumption is factored into the interpretation of the data and the effect on the result.	

					identify any bias it might introduce into the conclusions.	

Data

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
6. Quality of data collected (plots/tables/ inline)	No plots or data tables.	Cannot determine the quality of the data. E.g., points are not clear, or captions do not properly label the data so it is not clear where it has come from or what is supposed to be represented.		Low-quality data: data that is under-resolved or clearly does not show what is being claimed to be measured. Some quantities are missing uncertainties.		Medium-quality data: some, but not all, data meets the criteria of high-quality data. Uncertainties are provided on all measured quantities.		High-quality data: data that is well-resolved given instrument/ methodological limitations. Plots include error bars where appropriate. Plots clearly indicate the trend/relationship that is being claimed in the main text. Key results can be determined from the figures and captions alone.	

Conclusions

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
7. Conclusions	There are no conclusions.	Conclusions do not answer the research question or aim.		Conclusions are **not** based on fully correct physics but do provide a quantified response to the RQs. OR Conclusions are based on correct physics but do **not** provide a **quantified** response to the RQs.		Conclusions are based on **valid physics** and link back and provide a quantified response to the research question.		+ Conclusions provide valuable insights to the reader into the physics of the system being studied.	

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
8. Evaluation of conclusions through uncertainty analysis	There is no uncertainty analysis or evaluation of the conclusions.	A qualitative evaluation of the conclusions is provided with little supporting evidence. Uncertainties on the final result have not been considered.		The uncertainties in the final measured/calculated values are used in the evaluation of the conclusions. These uncertainties are not entirely consistent with the data and analysis presented.		Conclusions are justified based on the data and analysis presented. Implications of the uncertainty associated with the results on the validity of the conclusions are considered.		Conclusions are justified based on the data and analysis presented. The **size** of effects/differences and <u>appropriate</u> tests of the **statistical significance** of results are calculated and the implications of quantified systematic and random uncertainties arising from the methods chosen on the validity of the results is discussed.		+ Conclusions are consistent with **all** statements made and the limitations of the experiment. The answer to the RQ is articulated in terms of a hypothesis that has been tested and an <u>appropriate</u> method (relevant to the practice of the field of physics) of testing that hypothesis is used.	